SUBMERGED WETLANDS

A speculative design proposal. To reduce cyanobacterial bloom in Mälaren.

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Abstract

In the Anthropocene era, human intervention has significantly transformed our environment, blurring the lines between the natural and the artificial. This project places an emphasis on this viewpoint in environmental philosophy, and its role in the practice of the industrial designer. Drawing on research of two science advisors and their findings, the focus of Submerged Wetlands is to create a small-scale design of an artificial wetland, that works to mitigate cyanobacterial blooms, a growing phenomena caused by climate change,
specifically agricultural nutrient runoff and global warming.

The material choices, the plant Juncus Repens as an acting low-tech filter to remove excess nutrients, the choice of reed grass and the consideration of what people and space the design interacts with, comes out of natural fieldwork, interviews, lab tests and excursions to institutions.

In *submerged wetlands*, the properties of marinal vegetation are being speculatively utilized in a design proposal to construct submerged gardens under buoys in recreational areas around Lake Mälaren in Stockholm. These areas are subject to cyanobacterial blooms, commonly referred to as algae blooms, which pose a threat to humans and aquatic species.

Together with a body of scientific research on wetlands, soil and cyanobacterial blooms, the designed solution *Submerged Wetlands* takes into account the spectator's need for a third place, their emotional disposition towards the climate crisis and presents a demonstrative and accessible formation in the form of an artificial wetland.



Image 1. Close up of the submerged wetland

Introduction

This project stems from a critical stand point, where I as a designer want to question the authenticity of my surroundings. By authenticity I mean what the materials, interventions or manipulations and processes are. How ”organic” is my material reality? I realise that everything surrounding me is artificial. It is artificial in the sense that everything is human made or manipulated. Even my cat, Nelson, is the byproduct of generations of human intervention and breeding techniques. In less general terms, I wanted to apply the conflict between the authenthic and the artificial on my own process and decision-making in the graduation project.

The practice of an industrial designer, which in simple terms is a creative and constructive role, has to take into account the natural crisis we are currently in. Designers do this by identifying specific environmental problems, by to the best of their ability informing themselves factually, and in some instances by working with experimental materials to produce products in a different, biologically benign direction.

As someone who has studied environmental science in the past, I have long felt this mindset feels naïve and short coming, because these well-intentioned attempts are isolated, singular and for the few: they still exist in a larger production system that is problematic. I picture everything in my environment that is artificial, and I recognize that we also need to change the structures that we have built for ourselves in our practice and which we reinforce. We need to inconvenience our lives to save them.

In short, and based on the definition from Paul Crotzen who credited to have established the term, the Antroposcene describes a specific age and time of human intervention from a geological standpoint (Britannica, 2023). This is the period where we started to domesticate, dominate, exploit and alter our environment and the species around us.

The choice of wetlands as both a material base and as a natural concept, is partly a ”recent” issue (recent as in an event that has received more and more media coverage lately), making it accessible or known. It is also partly because it was a fitting and limited enough ground for experimentation.

With this specific biologic phenomenon, I could apply the contradiction of the artificial and authentic, the criticism of the practice of an industrial designer and learnings or outcomes of trying to give form to a solution, onto a natural and existing occurance. Wetlands as a natural occurance and problem area, motivated my work with unconventional materials such as reed grass and Juncus Repens.

Taking a position outside of the market and industry ”way of seeing” is part of my critical approach, and consulting with scientists is also a conscious way of recognizing expertise elsewhere than among commercial actors, that may or may not reinforce a different perspective of an industrial designer and their role. The progress of the project was dependent on recommendations and direction in the process of material experimentation and designing, from a scientific view, which gives it credibility and substance.

In essence, *Submerged Wetlands* takes the role of the designer into question, tries a scientific viewpoint and positions the maker and observer outside of commercial terms and takes the learnings from material experiments based on science to give form to a constructive suggestion for a solution that tackles a specific environmental problem, the rise of Cyanobacterial bloom.

Background

Anthroprocene Viewpoint

In the first sentences of the chapter *"Preface to a Genealogy of the Post natural"* out of *Land & Animal & Nonanimal* (W. Pell & Lauren B. Allen 2014), the reader is introduced to terms such as deforestation, extinction and atmospheric alterations. They are all linked to one event, and it is the Anthropocene. The project Submerged Wetlands is an attempt at approaching an environmental issue from the Anthropocentric point of view. It was important for the work to include advisors that hold this viewpoint and that could inform material and design choices with scientific reasoning from this position.

Earth and Soil

Given the direction this project was going to take, towards a complex and dense area of expertise, I aimed sources or bibliography to limit itself to metatexts or essays that would summarize and exemplify ongoing environmental challenges. Sources such as *Commons in Design* (2023) and *Land & Animal & Nonanimal* (2014) are anthologies of sorts, that detail process and method behind industrial design projects or popular ecology that are tangible to me, in the approach, the problem description and undertaking: it can be difficult to not become both the designer and scientist in a project as this.

In this period of resarching, I came across one natural crisis referenced, that is often overlooked, which is the degradation of our soil. In short, the foundation of human lives relies on soil and according to the *soil association*, 95% of our food comes from crops that naturally rely on soil to grow (Soil Association, 2023). Soil also plays a role in regulation water cycles, sustaining ecosystems and provides a home for countless organisms.
In our historical shift to pursue industrial agriculture and urban expansion, we have depleted this precious resource to extreme levels of degradation. The soil crisis is manifested in various ways, a few being erosion and compaction, commination and depletion of nutrients. Industrial farming practises rely on planting the same crop for multiple seasons, and an excessive use of pesticides that is stripping the soil of its fertility, rendering it lifeless. In simple terms, the problematic practice ends up leaving vibrant ecosystems in a void of non-existent biodiversity (ISRIC, 1991).

The consequences of a depleted soil reach further and after effects becomes more complex, as it does not only impact agricultural productivity. The runoff pollutes waterways and threaten marine ecosystems and leads to algae bloom in our oceans and lakes. Depleted soil does not sustain life and yield less crop. This perpetuates a cycle of poverty and hunger that impact communities globally.

To return to the initial view of this epoch, the Anthroprocene, it is a clear example of exploitative and altering behavior with a proven measured negative effect on its enviroment. From the perspective of a designer, the choice of how to engage with an issue as such becomes a material study, an exercise in navigating faculty language and most importantly, with the premise in the back of my head, how to sculpture something that is non-invasive, has longievity, is accessible and traceable. Any intervention or artificial additive would have to justify itself and not enable an existing dysfunction or contribute to more degragation.

Principles of Practice

I have just detailed some large issues: the fate of soil, the Anthropocene, and a shift in agricultural production. The soil degradation issue quickly became overwhelming for this reason, in terms of pin-pointing something formative and actionable. In parallell, I also found tangent societal changes that I felt, on an empathetic and receptive level would be important to follow as principles in a project that could become distantly scientific or difficult to take to heart.

Throughout my studies, I have kept returning to the sociological concept of third places when creating work. The first place being home, the second being a place of work, Ray Oldenburg explains the third place as “a space

for informal, free social interaction, essential to democracy" (Oldenburg, 1989). Increasingly, with the commercialisation of public space and move towards virtual mediums, forming informal spaces is key to my work.
To exemplify, in the publication *Ett södermalm som gör dig varm* (Holfve 1985) is a set of stories from Stockholm burrough Södermalm and a collection of personal contact ads and local advertisements of the time. Common words used in this book are gemenskap (togetherness), folkligt (for the people) and sällskap (company). These words are based in vernacular ideals and values that forwards collectivism, equality and public space. On an anectdotal level, my sense was that the third place had its final shift to the internet as a result of the fast social digital development in the pandemic, and that we are now accustomed to being deprived of physical or material third places. The need to organize around a cause, join a community or find common practice in a hobby felt unmet.

Activation Points

Part of this activation is where to direct feelings of hopelessness, distrought and overwhelming doom when being made increasingly aware of the current environmental crisis and future ecological catastrophes to come. Having this sense myself, I reached out to organization "Klimatpsykologerna", which is profiled as dealing with patients mentally affected by climate changes. The syndrome, climate anxiety, is defined as "*heightened emotional, mental or somatic distress in response to dangerous changes in the climate system*" (NCBI, 2021). My request to "Klimatpsykologerna" was to interview an employee about their process and methods in treating climate anxiety. My correspondence with Klimatpsykologerna was unsuccessful, after the organization required me to submit a fee of 1100 kr for a session, but was unwilling to take part in an interview or share their practice. It felt concerning to be met with indifference or disinterest in sharing or giving insight into their organization. From a critical perspective, their operation relies on profiting from an emerging mental issue among a significant group of people, and it would have been credible and inspiring for them to help inform the scale of the problem, the tools at hand to solve it and the results so far. This would likely have helped steer the psychology of the project (what emotional aspects would go into the design). However, from this failed attempt at including a second party, I took it that likely there is a significant group of people that has tried to find a third place or a community of others who share their sentiment, specifically climate anxiety and this need to act on distressing information, or place an awareness onto a specific cause. With the third place as an abstract ideal in mind for the form and elements of my installation, with the experience of a failed outreach to an organization profiting on environmental issues, I had meaningful motivations to base the communication and form of my work around.

Giving Form To Science

Having read and digested articles, essays and lectures on the topic, I decided I needed to externalize the project. This moment of the research process was crucial: when you are out of your element and also when your project becomes limited to the context it is made (Konstfack), it will not live up to its potential in substance and execution. Being a former student of sustainable development, my concepts and interest often stem from natural phenomena, and I turn to scientists for artistic wayfinding. This had me contact Yrsa Larsson, a PhD student in the field of soil in the Biochemistry Department at Aarhus University, and we connected virtually.

After having explained the intent or spirit of the project, forwarding the research and my questions, she explained to me that her field of research is about antibiotics and how it creates multi-resistant bacteria. She pointed me to other relevant material and suggested to contact her colleague. Besides the insight into her scientific practice and areas of research, what the correspondence with she highlighted was the need for someone outside of her field of expertise to properly communicate or give form to research that often does not reach the public. More than the substance of what we discussed in her research, this is where I found my purpose: using her existing research as a foundation for my project and have it guide and inform my choices.

Cyanobacterial Bloom

The collegue of Y. Larsson is another PhD student, Alba Martinez I Quer. She is a co-author of a study on *cyanobacterial blooms in surface water reservoirs, Nature-based solutions for cyanotoxin remediation* (Y. Larsson et al. 2024). Alba filled me in on her undertaking of cleaning water that has been contaminated by manure, where she creates and uses artificial wetlands as a cleansing method. The plants used to grow an artificial wetland are Phragmites, a type of reed grass common in wetlands — and Juncus Effusus, a rush that grows partially submerged in water. When reviewing lab footage, most noticeable was a photograph that depicted two water filled vessels: one with black water and one with clear water. It was explained to me that the black water vessel was contaminated with manure. The clear water vessel was the same liquid, only after being pumped trough the plants rendering it clear as a result. It was a demonstrative and relieving image for an outside spectator, to see a scientific solution designed and applied to make a visible difference. This was what I was hoping to accomplish with my own design solution: to use visual cues, material choices and a clear process to guide the spectator. I learned the properties of these plants, the reed grass and rush: being ideal for absorbing nitrogen and phosphorous, being the main ingredients in eutrophication and causing cyanobacterial bloom in oceans and lakes. These would become material sources of mine, but specific to their context: the wetlands. The wetlands as a contested area, involving both expoloitative and preservative forces in conflict, is a talking point in environmental politics: aside from providing a habitat for many different kinds of species, reducing erosion, filtering water and functioning as natural carbon sinks, I viewed them as commonplace and relatable as symbolic spaces for environmental issues and locally relatable climate change.

In an empathetic way, the loss of the third space and the growing climate anxiety, could both be engaged in this space. Together with the excursions and correspondence with the scientists, their material findings and laboratory tests to a natural environment there was also a throughline with my extensive research on soil degradation: the specific phenomena of nutrient runoff that causes an overgrowth of cyonobacterial bloom, became a sizeable issue to try and formulate a solution for, which was already based in the research of Y. Larsson and A. Martinez I Quer.

Purpose

A large part of this project's learning outcomes, was to apply a critical approach of who I work with, what information sources I base my material choices and for whom I design, what their needs are and will be in the future. This means to take a position in practice, that is contrarian to market conditions. To be more specific, this means to apply design thinking and problem formulations that I have learned in my studies for the role of an industrial designer in a environmentally conscious and socially sustainable way. This is in contrast to adapting decisions and production in the same production design behavior, which environmental philosopher Glenn Albrecht describes as "… growth-addicted industrial-technological society that has achieved its success at the expense of the vitality of the Earth." (Albrecht, 2016). To continue on what is mentioned in the background to this reader, I've also wanted to make these decisions specifically conscious of what is artificial, what is a harmful and what is a helpful human intervention into a natural occurrence, and how I can apply these values in actions, design choices and material decisions within the process of the project.

Another important motivator of the project is to manage to communicate, via formation, exhibition setup and process documentation, how an industrial designer can take on a complex natural phenomena, often only fully understood and discussed among scientists, and make it tangible and communicative in my design solution. The choice to work with wetlands, comes partly out of extensive research on the current soil degradation crisis, and it's multilayered after effects: it's impact on agriculture practices, on global food supply and on water based eco-systems. The connection between the process of soil degradation and the state of wetlands is spelled soil. Wetlands are interconnected with soil and agriculture: where groundwater meets the soil surface, there is a wetland, in simple terms, although their placement and wetness may vary.

For clarity, by wetlands, I refer specifically to those accessible in my vicinity, in Västerjung and Stockholm, which mostly are inland and non-tidal Wetlands, also known as marshes and wet meadows (EPA, 2024). There are several types of wetlands including marshes, swamps, bogs and fens. According to Naturvårdsverket, wetlands are areas where there is a presence of water, the soil must be saturated in water either seasonally or permanently. When the areas are dry the area is qualified as a wetland if more than 50% of the vegetation is a hydrophytic plant, which means that the vegetation is adapted to thrive in waterlogged soil. These factors makes wetlands vital for biodiversity, water filtration and flood control. The purpose of working with inland wetlands was to be able to do fieldwork, meaning visiting natural sites and wetlands accessible from my home or country house and engaging with the environment I study, problematize and design for.

Partly, the decision of wetlands as an area of study also lies in the limitations of working with the research specific to the scientific advisors involved in the project, their knowledge and direction. Both scientists, Y. Larsson and A. Martinez I Quer provide the body of research on how nutrient runoff from agricultural soil causes cyanobacterial bloom — and how the usage of artificial wetlands can prevent this. Lastly, the scope of working with wetlands as a specific eco-system, with its own unique properties, conditions and needs, is a needed limitation to apply the intentions behind this project, and can at the same time function as a symbolic or scalable environmental challenge on which the industrial designer can apply herself.

The study of the properties and elements of wetlands, the type of vegetation and species animals present, is a stepping stone to make conscious interpretations — some artificial and some authentic. It informs the making of an installation, a product or design, that simulates a wetland and the same mending processess that wetlands have on soil degradation, only artificially produced without harmful human intervention. The specific purpose of Submerged Wetlands as a design solution, is to act as this natural filter in the same demonstrative and clear way as in A. Martinez I Quer's research fin-

dings (the laboratory review photography of water vessels).

The artificial wetlands I have designed are shown as a scientifically sound solution, capable of removing nitrogen and phosphorus from Mälaren — a lake covering vast areas of Stockholm with many recreational areas for water activity and a rich biodiverse eco-system. Mälaren was purposefully chosen for Submerged Wetlands as the environment to design for: it is annually, in the warmer seasons, often struck by cyanobacterial bloom (commonly known as algae bloom), which an indirect consequence of soil degradation. According to Mälarens Vattenvårdsförbund (n.d.) the occurrence of cyanobacterial bloom is most common during the middle of summer, when the winds are calm. The toxins is dangerous to ingest, does not go away from boiling and leads to deceased flocks of animals. The structure of the design, how the reed grass is tied and supported by metal hangers, provide soil filled areas that hold the plant Juncus Repens. These decisions are reflections of how wetlands behave naturally, according to the research of my advisors: specifically how the plants are placed and for what purpose.

The interventions were going to be taking into account the empathetic and social aspects of my practice: the need for a tangible, clear example of an Anthroproscene design that does not negatively effect or interrupt existing conditions of the environment it is placed in was the first purpose. For anyone to take in and comprehend the project, I my intention was also to design something spatial, interactive and traceable. Spatial in the sense that it is accessible, recognizable and communicative in its features; interactive in the meaning that a viewer can understand both by form and placement what each part in the design does for its entirety. Partly by using visual cues (such as imitating nature or working with materials they've been exposed to) and partly by being decisive of how things are arranged. Lastly, the traceability does not mean the material origins, which are also important, but more so the documentation and demonstration of the process behind the work. These intentions had everything to do with the reception of the work, but the themes of saving a third place and the experienced climate anxiety are both present and woven into these decisions as well.

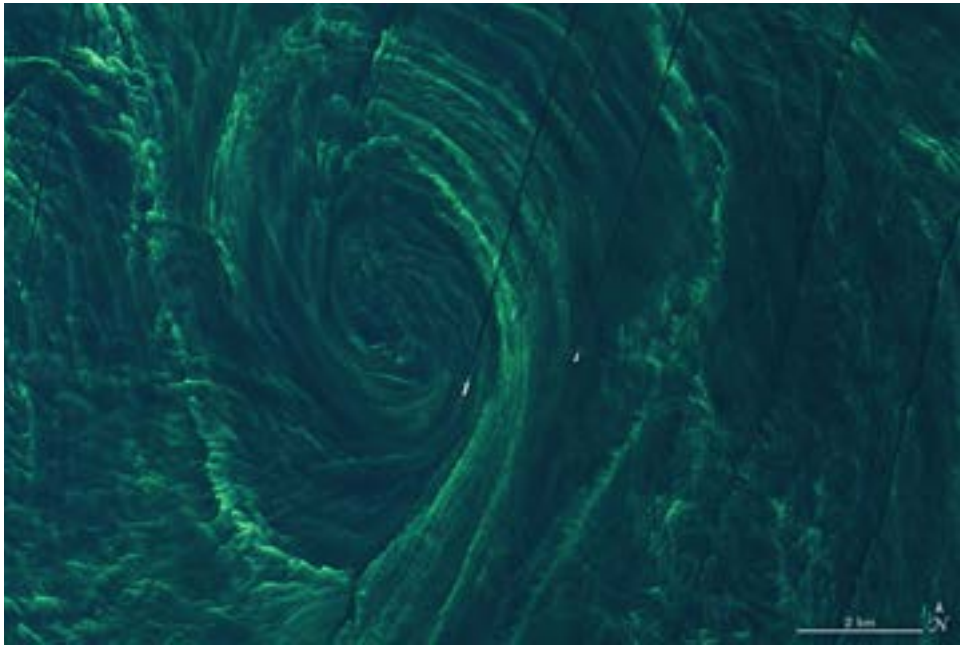


Image 2. Cyanobacterial Bloom (NASA Earth Observatory, 2020)

Design Process

Understanding wetlands

In order to build a wetland, it was crucial to fully understand what elements constitute it. For this reason, I contacted Gunvor Larsson, the garden superintendent at Bergianska trädgården, a botanical garden affiliated with Stockholm University. We met in a large yellow villa near the botanical garden, which is adjacent to the area's wetlands. At this point, the feeling of being a designer overreaching to become a scientist struck me: I was attempting to build a small scale wetland from scratch, but I didn't know where and in which end to begin. In her office, talking to Gunvor, I was comforted by learning what I needed to know in order to get started. She began with describing how the wetland came to be at Bergianska. This she did by showing me maps of the area that is now a wetland, and how it had changed during periods of time. It is speculated that the area was a wetland to begin with, but that it was drained during the 19th centuary, not unlike like many other wetlands did in that period in time. The wetland then became a hayfield and changed into different types of farmland up until 2009. In 2009 Bergianska decided to transform the land into a wetland once again. The wetland has shifted trough different "forms" of wetlands. In the beginning they had an issue, where there was too much reed grass in the area. This posed a problem, because reed grass can grow aggressively fast compared to the other plants and therefore overrun the other plant species. Even though Y. Larsson and A. Martinez I Quer use phragmites in their artificial wetlands, the intervention can cause issues in the natural world. Gunvor showed maps of how the landscape has changed during the centuries and also showed me how they managed to build it in 2009. Wetlands are ecosystems where water saturates the soil, either permanently or seasonally, supporting aquatic plants and providing essential ecological services. To qualify as a wetland, an area must meet specific hydrological, soil, and vegetation criteria. This project focuses on artificial wetlands that mimic these natural systems to purify water by absorbing excess nutrients.

Historical draining of wetlands

Länstyrelsen Västra Götaland (2008) describes how the landscape shifted in Sweden during the mid-eighteenhundreds and forward. Initially large areas were permanently or periodically wet but as agriculture became more important for Swedish society drainage of wetlands became a common practice to make way for farmland. Forest drainage followed a similar pattern. It halted during World War I, but resumed during the 1930s economic crisis, which among other phenomena, made clear the exploitative nature of our econonic system and its impact on specific natural environments.



Image 3. Bergianska Trädgården wetland

Visiting wetlands

I visited Vagnhärad våtmark to take some tests to measure how the wetlands change the properties of water. I filled a small bucket with a sample of the water, plants and sediment. Because the season, being early spring, there were no significant signs of natural life yet. The wetland in Vagnhärad is connected to a river and water runs trough the wetland. The water running in before the wetland was more acidic, than when it was in the middle of (running through) the wetland. The water became more alkaline in the end. Acidic water is often not a fertile condition for biodiversity, and appears as a byproduct of increased carbon dioxide levels in the air that gets dissolved in the water. After Vagnhärad I went to Trosa våtmark to take some tests. However, this wetland was surrounded by a grid fence, followed by a sign signaling to visitors that it was marked a ”*protected site”,* due its environmental role as a cleaning agent for the water in the area Trosa Municipality (2024).



Image 5. Vagnhärad Våtmark



Image 4. Testing water in Vagnhärad Våtmark



Image 6. Trosa Våtmark sign saying “treatment of communal sewage water”

Building A Wetland

With the experience and learnings about wetlands from my field studies I made an attempt to construct a small scale environment similar to a wetland, for plants to thirve in. In order to do this properly, I needed to get my hands on the correct plants. Despite my efforts to source these plants from Vagnhärad våtmark, I was unable to find a suitable specimen.
After a run of visits to various aquarium stores in and around Stockholm, trying to acquire a sample of Juncus Effusus or Phragmites. Plants which A. Martinez I Quer details her work on, they proved hard to get a hold of for the average consumer. Having consulted with A. Martinez I Quer and Y. Larsson we found a close relative to Juncus Effus, Juncus Repens, which was accessible to a greater extent. The difference between the Juncus Effus and the Juncus Repens, is that the Repens is a smaller specimen, that likes to grow submerged in water in depths up to 1 meter. The replacement plant also grows naturally in wetlands, but not in Scandinavia. This is because it cannot thrive in water temperatures below 4°C, and so the climate and geographic location played a determinant role in this choice. I also concluded that the specific property of temperature tolerance was a positive aspect of the plant choice, as one could expect it to wither and decay in the winter, preventing it from potentially being an invasive species and making the intervention on the eco-system unharmful.

Juncus repens requires a solid substrate for its roots, which is explains my use of a mixture of gravel and sand at the bottom of the buckets, ensuring that both buckets contained the same weight of substrate. This laid the foundation for the formation of the prototype. Additionally, the aim was to research the cleaning properties of the plant.

I added manure to both buckets, but only one contained the Juncus Repens plant. After 17 hours, I observed a color change, and after 14 days, the water with the plant had become clear. Showing similar results as Martinez I Quers research with phragmites and Juncus Effusus. Due to a lack of a lab and scientist I could not determine the levels of phospherous and nitrogen being removed.



Image 7, Bucket experiment after 17h

The place of the submerged wetland

Mälaren plays a sentimental role to me. By its beaches and rockfaces, I have spent both my childhood and adulthood in different activities in around the ocean. During the summer, it becomes a well frequented area for engaging in various recreational water activities, increasingly trafficked by boats and ferries. This time of year, Mälaren also experiences a repeating annual and seasonal issue with cyanobacterial bloom. Accoarding to Tidningen Syre (2020), there is an app called CyanoAlert that warns and notifies visitors in and around Mälaren, when algae bloom is occuring, so they do not swim or enter the water.

I traveled to different recreational areas around Mälaren in Stockholm, to find a set of locations which environment and conditions were apt and well suited for the artificial wetland design. I decided to selecting locations close to where people swim, as it would be a contaminated area: and the layout of the land would have to be able to hold the construct up without adding more plastic. In order to test if samples I had made of bio-material would float, some tryouts were done at these locations. Only one of the specimens could endure the water, which was a type of bio material containing wood glue, which was the result I was after, since it contains plastic and could be harmful.

I saw that many areas where visitors are allowed to swim have a set of marker bouys placed in the water, to demonstrate where the simmers should not cross to avoid them getting into boat routes. I looked into the bouys floating capabilities and decided to include them in my design, by attaching them in their and placing the submerged wetlands under a set of them. I reached out to Stockholm Stad Hav & Vattenmyndighet, considering they regularly perform water tests and have authority over the swimming areas around Stockholm. They told me that they test the waters in Mälaren from May to September and the bouys are placed in between April and September.



Image 9. Initial sketch for how it would look and work

Image 8. Sketch on how it would look in Årstaviken



Material

To ensure the project was not introducing anything harmful to the water, the materials were carefully selected for the submerged wetland. The floating buoys at the recreational areas in Mälaren are already in place, and so the solution is based on their existing design and function. I purchased a buoy to determine the weight it would support, while still managing to remain afloat. The specific buoy I tested could withstand about 20 kilograms.

The holders for the plants needed to be cage-like in their design, to securely hold the gravel for the roots of Juncus repens. I decided to use reed grass because it is often cut down in wetlands to make way for other plants, according to Gunvor. Reed grass is a natural, renewable resource and has evolved to withstand being in water. Additionally, Thor Heyerdahl, a Norwegian explorer and anthropologist, once built an entire boat out of reed grass, showcasing its durability and strength.

For the structure situated under the buoys, I chose to work with aluminum, due to its light weight and high resistance to corrosion, ensuring durability and longevity. Aluminum is also non-toxic, making it a safe choice for aquatic environments. Although aluminum is recyclable, the production of new aluminum requires significant energy, so recycled aluminum is a preferable choice of material.

I constructed one version of the design completely out of reed grass and waxed thread to show a version one can make themselves and put in their own lakes if cyanobacterial bloom is occurring frequently.

Production and scenario

The production of the aluminum structures is intended to be done locally, and the structures will be reusable each season. Every year, reed grass nets will be placed on the structures. During the winter, the plants will overwinter at Bergianska Trädgården.

Each April, when the wetlands get attached to the bouys an event will be held. The wetlands are attached with a carabiner to the bouys. Where each participant can create their own reedgrass wetland to bring to their favorite lake.

This event aims to establish a future tradition that ensures more places are protected from Cyanobacterial blooms. The submerged wetland replicates natural wetland conditions by providing plants with an underwater structure and soil. In practice, extensive testing would be necessary to evaluate the amount of water purified and to determine the number of submerged wetlands required to alter phosphorus and nitrogen levels, particularly in a lake as expansive as Mälaren.



Image 10. Reed grass, and waxed thread



Image 11, The prototype of the submerged wetland

Inspiration and Formation

Taking inspiration from the wetlands I visited, I decided to look up satellite images on Google to examine their shapes. I had been exploring the wetlands from the perspective of a human, but I wanted to imagine the viewpoint of a bird. Many wetlands I visited harbored beautiful birds, and I have always been jealous of their ability to fly away, soar through the skies, and look down on the world.

The satellite images of the wetlands were stunning, and I fell in love with their soft outlines. Inspired by these shapes, I began drawing them and shaping metal into these forms. This approach allowed me to capture the essence of the wetlands' beauty and their fluid, organic patterns.

For the structure completely out of reed grass, I was inspired by bookbinding techniques and explored how I could make different shapes using the sticks and making organic formations that could hold the gravel.



Image 12, Prototyping with reed grass and thread



Image 13. Prototype with Juncus Repens before sewn together



Image 14. Sketching the shapes of the wetlands



Image 15. Metal structure with shapes of wetlands

Design Proposal

Submerged Wetlands aims tackle the growing problem of cyanobacterial bloom in Mälaren, Stockholm. Cyanobacterial blooms, fueled by climate change and agricultural runoff, pose serious health risks to humans and aquatic life by depleting oxygen and reducing biodiversity. The solution involves using the plant Juncus Repens as a natural filter to remove excess nutrients from the water.

The project has three main goals: improving environmental health, engaging the community, and using sustainable materials. using the existing floating buoys in recreational areas to support the submerged wetland structures. The plant holders, made from reed grass and aluminum, will securely hold the plants and gravel.

Juncus Repens, chosen for its effectiveness in nutrient filtration, will be planted in these holders. The wetlands will be installed each April and removed in September. During winter, the plants will be stored safely at Bergianska Trädgården. In spring a community event will be held when the wetlands are connected to the bouys where people can make their own reedgrass wetlands and place them in a chosen area.

In Mälaren the submerged wetlands is encouraging visitors to peer beneath the surface to discover the sunken artwork. The installation not only provides an aesthetic experience for humans but also enhances the habitat for aquatic life, creating a harmonious blend of art and nature. Even the fishes has something to look forward to in spring.



Image 16. Photo from exhibition



Image 17. Reed grass structure with Juncus Repens



Image 18. Aquarium with reed grass structure



Image 19. Table at the exhibittion, the two tubes cointaining the water before and after Juncus Repens cleaning it



Image 20. Reed grass structure for the open source version

Reflection

This project has taught me a lot about wetlands and at times it felt like I needed to turn every rock to make it into something. If I had had more time I would have became a scientist and then made this happen in real life, testing it in water.

It was a dream come trough to work with so many intresting people during this project. I would have liked to talk to more people and making this a reality by implementing this in Mälaren turning it into an tradition.

In the future I want to focus more on the formgiving aspect and see how people interact with such a thing. Working more on how the bouys and having educational signs. I learned that working with certain plants is difficult during an off season and the project might have been different if I had other plants as well.

It was so much fun too show it at the exhibition and I got very nice responses from people seeing the project, both from adults and children. So many people shared that they wished it was implemented already, and wanting to have one at the places where they go swimming. I hope to make that a reality, for both human and other species.

I would also like to see what would happen with other species as well, getting to know the aquatic friends and how they interact with it. I am proud of myself and how I implemented the research into the project. I want to continue exploring the Antropocene.

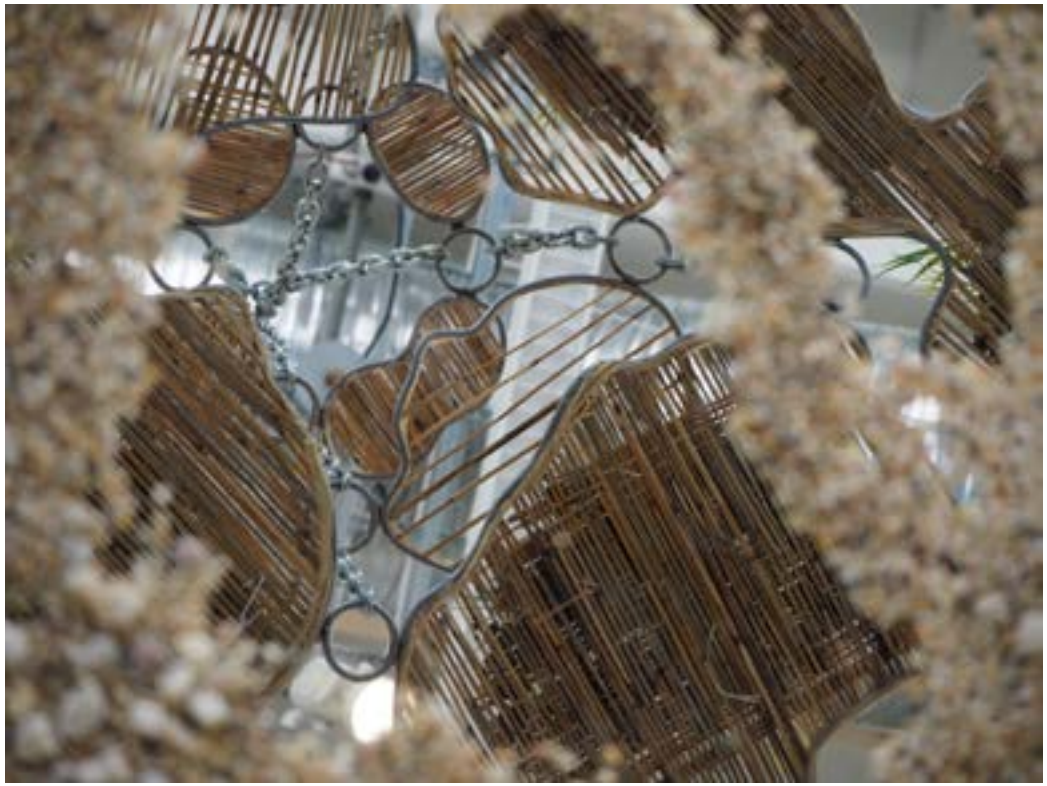


Image 21. The reflections from the mirror on the floor

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image 1. private photo

image 2. NASA Earth Observatory (2020) Beguiling Bloom in The Baltic Sea. Available at: https://earthobservatory.nasa.gov/images/147135/beguiling-bloom-in-the-baltic-sea (Accessed: 28 May 2024).

image 3. private photo

image 4. private photo

image 5. private photo

image 6. private photo

image 7. private photo

iamge 8. sketch over photo

image 9. sketch

image 10. private photo

image 11. private photo

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image 14. sketch over google satelite image

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